

The Lacunarity Boundary

Two-Moment Rigidity and Sharp Recovery of Geometric Uniform Spectra

Resolving the critical dyadic separation question,
with sharp prefix conditioning and quartic testing complexity

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Abstract

Let $X_a = \sum_{j \geq 1} a_j U_j$, where the U_j are independent uniforms on $[-1, 1]$. We resolve the exact separation-boundary question posed in the ProveIt article *Anchored Recovery of the Dyadic Uniform Spectrum*: under $a_{j+1} \leq a_j/2$ and $\sum a_j^2 = 1/3$, the full coefficient sequence is recoverable from its law with the sharp exponent $1/2$. In fact, the fourth moment alone gives the global certificate

$$\sup_j |a_j - 2^{-j}|^2 \leq \frac{75}{4} \left(\frac{19}{675} - \mathbb{E} X_a^4 \right).$$

The expression in parentheses is nonnegative and vanishes only at the exact dyadic spectrum. Thus two moments identify an infinite spectrum inside this one-sided constraint class. A deleted-factor construction proves matching square-root lower bounds in total variation and Kolmogorov distance, even inside the predecessor's precise infinitely smooth class. Fixed-prefix local Lipschitz constants grow as $\Theta(2^r)$, and testing the reference against spectral distance at least δ requires and suffices to use $\Theta(\delta^{-4})$ observations. The statistical lower bound uses a boundary-sensitive Hellinger estimate, not a product-total-variation shortcut. We also prove exact support rigidity, extend the method to every geometric reference and to nongeometric relative-ratio cones, and separate these results from the still-open pairwise and small-slack problems.

Status and scope. The new statements are proved in this manuscript, but have not been checked in Lean or independently peer reviewed. The claim of novelty is a proposed repository-relative research contribution, not an exhaustive literature-priority determination. The basic summation-by-parts, convolution, and concentration tools are classical.

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1 The repository question and the main conclusions

1.1 Why the exact boundary is a different inverse problem

The ProveIt Fabius-function collection contains the research article *Anchored Recovery of the Dyadic Uniform Spectrum* [1]. Its third further-research question asks what happens under the exact inequality $a_{j+1} \leq a_j/2$, with variance fixed at $1/9$. That article constructs very small law perturbations when any fixed slack above $1/2$ is available, but explicitly leaves the boundary case unresolved. Its construction preserves many power sums by compensating changes of neighboring scales; the exact one-sided constraint obstructs those changes.

The present answer is positive and considerably more explicit than a qualitative inverse-continuity statement. At the boundary, the fourth moment is an exposing functional for the reference spectrum: its deficit controls every squared coefficient. This reduces an infinite-dimensional anchored inverse problem to one scalar measurement. It does *not* say that two moments determine all spectra, or that two freely varying spectra admit the same inverse estimate. The reference is the unique moment extremizer inside the specified cone.

For comparison, the predecessor proves the anchored scale $\exp[-\Theta(\sqrt{\log(1/\varepsilon)})]$ on its smooth classes with every fixed ratio bound $\rho > 1/2$ [1]. Our critical-boundary proofs do not import that unformalized quantitative result. The comparison with it is an attributed repository result; the positive and sharp boundary theorems below are established independently.

1.2 The dyadic statement

Throughout, $U_1, U_2, \dots$ are independent random variables uniform on $[-1, 1]$. Put

$$a_j^0 = 2^{-j}, \quad X_0 = \sum_{j \geq 1} 2^{-j} U_j, \quad \mu_0 = \text{Law}(X_0). \quad (1.1)$$

The density of μ_0 is the centered Rvachev up-function, here normalized on $[-1, 1]$. This probability interpretation and the compactly supported smooth function are established background [2]; we derive the moment and norm formulas needed below directly.

Definition 1.1 (Critical dyadic class). Let

$$\mathcal{C}_* = \left\{ a = (a_j)_{j \geq 1} : a_j \geq 0, \quad a_{j+1} \leq \frac{1}{2} a_j, \quad \sum_{j \geq 1} a_j^2 = \frac{1}{3} \right\}. \quad (1.2)$$

For $a \in \mathcal{C}_*$ write $X_a = \sum a_j U_j$, $\mu_a = \text{Law}(X_a)$, $m_k(a) = \mathbb{E} X_a^k$, and

$$d_\infty(a, b) = \sup_j |a_j - b_j|, \quad d_r(a, b) = \max_{1 \leq j \leq r} |a_j - b_j|.$$

Finite sequences are padded by zeros. All such random series converge absolutely because $a_j \leq a_1 2^{1-j}$.

Our total variation convention is $\text{TV}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$, one-half the L^1 distance for densities. Kolmogorov distance is $d_K(\mu, \nu) = \sup_x |F_\mu(x) - F_\nu(x)|$. In particular $d_K \leq \text{TV}$.

Theorem 1.2 (Critical dyadic recovery). *Every $a \in \mathcal{C}_*$ is supported probabilistically in $[-1, 1]$, has $m_2(a) = 1/9$, and satisfies*

$$e_4(a) := \frac{19}{675} - m_4(a) \geq 0, \quad (1.3)$$

$$d_\infty(a, a^0)^2 \leq \frac{75}{4} e_4(a), \quad (1.4)$$

$$d_r(a, a^0) \leq \frac{75}{4} 2^r e_4(a) \quad (r \geq 1). \quad (1.5)$$

Equality $e_4(a) = 0$ holds if and only if $a = a^0$. Consequently,

$$d_\infty(a, a^0) \leq \sqrt{\frac{75}{4} \text{TV}(\mu_a, \mu_0)}, \quad (1.6)$$

$$d_\infty(a, a^0) \leq \sqrt{\frac{75}{2} d_K(\mu_a, \mu_0)}. \quad (1.7)$$

The exponent $1/2$ is optimal for both distances, even after imposing the exact smooth-class conditions in theorem 4.2.

The two distinct scales in (1.4) and (1.5) are important. An error concentrated at a coefficient of size h changes its square by order h^2 , so a global loss naturally has a square root. A fixed coefficient is bounded away from zero, so dividing a squared-coefficient error by that coefficient gives a Lipschitz bound. We prove that its dependence on the prefix length is also sharp.

Theorem 1.3 (Statistical and conditioning consequences). *At the dyadic reference, the local Lipschitz constant for the first r coefficients is $\Theta(2^r)$ in either law distance. Testing $a = a^0$ against $a \in \mathcal{C}_*$ with $d_\infty(a, a^0) \geq \delta$, with both error probabilities at most $1/4$, has sample complexity $\Theta(\delta^{-4})$. Both lower bounds remain valid in the exact smooth subclass. An honest moment-based confidence set over $\mathcal{C}_*$ contracts at a^0 with radius $O(N^{-1/4})$, and its fixed-prefix radius is $O(2^r N^{-1/2})$.*

These are anchored testing and confidence statements. In particular, no claim is made about a globally optimal estimator between arbitrary spectra. At a single known point the constant estimator has zero loss, so a bare “pointwise minimax rate” would not express the content of the testing theorem.

1.3 Established background and the proposed contribution

Billey and Swanson study generalized uniform sums and their unique sequence-space parametrization, including a topology allowing Gaussian components [3]. Their qualitative identifiability is not a new claim here. Arias de Reyna develops the smooth compactly supported dyadic function and its probability and derivative representations [2]. Hoeffding’s bounded-variable inequality supplies a standard concentration input [4]; a short proof of the version used here is included.

The proposed contribution is the quantitative rigidity at the exact one-sided ratio boundary, together with smooth sharpness witnesses, sharp prefix conditioning, Hellinger-controlled testing complexity, and the relative-ratio extension. The algebraic core is elementary enough to isolate for formalization. Elementary tools do not make the inverse conclusion automatic: the square-slack bound, preservation of the predecessor’s infinitely many smoothness inequalities, and the fourth-order Hellinger estimate each address a separate obstruction.

2 A geometric majorization identity

Fix $0 < \rho < 1$, write $q = \rho^2$, and fix $S > 0$. Define

$$\mathcal{C}_{\rho, S} = \left\{ a_j \geq 0 : a_{j+1} \leq \rho a_j, \sum a_j^2 = S \right\}, \quad g_j = \sqrt{S(1-q)} \rho^{j-1}. \quad (2.1)$$

Thus g is the geometric reference with exactly the same ratio as the constraint. When $\rho > 1/2$, this reference is *not* the dyadic reference; our general theorem therefore does not conflict with the predecessor’s instability at the dyadic anchor under a looser ratio bound.

Set

$$p_j = \frac{a_j^2}{S}, \quad b_j = (1-q)q^{j-1}, \quad c_q = \frac{1-q}{1+q}, \quad \Delta = \sum_j p_j^2 - c_q, \quad d = p_1 - (1-q). \quad (2.2)$$

Both p and b are probability sequences and $p_{j+1} \leq qp_j$.

Lemma 2.1 (Majorization with an exact remainder). *For every probability sequence p satisfying $p_{j+1} \leq qp_j$, put $D_k = \sum_{j \leq k} (p_j - b_j)$. Then $D_k \geq 0$ and*

$$\sum_j p_j^2 - c_q = \sum_j (p_j - b_j)^2 + 2(1-q)^2 \sum_{k \geq 1} q^{k-1} D_k. \quad (2.3)$$

In particular $\sum p_j^2 \geq c_q$, with equality only for $p = b$.

Proof. Iterating the ratio inequality gives $p_{j+k} \leq q^k p_j$. Summing over $j \geq 1$ yields $\sum_{j > k} p_j \leq q^k$. Therefore

$$D_k = q^k - \sum_{j > k} p_j \geq 0.$$

Since $\sum b_j^2 = c_q$, expanding the square gives

$$\sum p_j^2 - c_q = \sum (p_j - b_j)^2 + 2 \sum b_j (p_j - b_j).$$

Finite summation by parts, followed by a limit, transforms the last sum into $\sum_{k \geq 1} (b_k - b_{k+1}) D_k$. The boundary term is $b_N D_N \rightarrow 0$, because D_N is bounded and $b_N \rightarrow 0$. All series converge absolutely; in particular $0 \leq D_k \leq 1$ and the weights are geometric. Finally $b_k - b_{k+1} = (1-q)^2 q^{k-1}$, proving the identity. If its left side vanishes, the squared-error term is zero coordinatewise. $\square$

The next refinement is stronger than discarding the nonnegative remainder in (2.3). A direct ℓ^2 estimate followed by taking square roots would lose an unnecessary additional square root. We instead first control the head excess and then all envelope slacks.

Lemma 2.2 (Sharp head-excess inequality). *With the notation (2.2), $0 \leq d \leq q$ and*

$$\Delta \geq 2c_q d + \frac{2}{1+q} d^2 = \frac{2}{1+q} d(1-q+d). \quad (2.4)$$

This bound is sharp for every $d \in [0, q]$. Consequently

$$d \leq \mathcal{D}_q(\Delta) := \frac{\sqrt{(1-q)^2 + 2(1+q)\Delta} - (1-q)}{2} \leq \frac{\Delta}{2c_q}. \quad (2.5)$$

Proof. The bound $p_1 \geq 1-q$ follows either from $D_1 \geq 0$ or from $1 \leq p_1/(1-q)$. The remaining mass is $1 - p_1$. Apply theorem 2.1 to the tail normalized by this mass to obtain

$$\sum_{j \geq 2} p_j^2 \geq c_q (1 - p_1)^2.$$

When the mass is zero the same inequality is immediate. Substitution of $p_1 = 1 - q + d$ and expansion gives (2.4). Equality is attained by the sequence

$$p_1 = 1 - q + d, \quad p_j = (q - d)(1 - q)q^{j-2} \quad (j \geq 2).$$

Its first ratio satisfies $p_2 \leq qp_1$ precisely because $p_1 \geq 1 - q$; the later ratios equal q . Solving the quadratic proves the first bound in (2.5), and dropping its positive quadratic term proves the second. $\square$

Lemma 2.3 (Every squared coefficient is controlled by the head). *For $a \in \mathcal{C}_{\rho,S}$,*

$$|a_j^2 - g_j^2| \leq Sd \quad (j \geq 1). \quad (2.6)$$

Proof. Define the nonnegative envelope slack $R_j = Sp_1 q^{j-1} - a_j^2$. The ratio constraint implies $R_{j+1} \geq qR_j$, and summation gives

$$\sum_j R_j = \frac{Sp_1}{1-q} - S = \frac{Sd}{1-q}.$$

On the other hand, $\sum_{k \geq 0} R_{j+k} \geq R_j/(1-q)$, so $0 \leq R_j \leq Sd$. Since

$$a_j^2 - g_j^2 = Sdq^{j-1} - R_j,$$

both terms on the right lie in $[0, Sd]$. Their difference has absolute value at most Sd . $\square$

3 Moment certificates, support rigidity, and global recovery

3.1 Moments of a uniform sum

Lemma 3.1. *For $a \in \mathcal{C}_{\rho,S}$, put $T = \sum a_j^4$. Then*

$$m_2(a) = \frac{S}{3}, \quad m_4(a) = \frac{S^2}{3} - \frac{2T}{15}. \quad (3.1)$$

For the geometric reference $T(g) = S^2 c_q$, and hence

$$e_4(a) := m_4(g) - m_4(a) = \frac{2S^2}{15} \Delta \geq 0. \quad (3.2)$$

Proof. For a centered uniform U , $\mathbb{E}U^2 = 1/3$ and $\mathbb{E}U^4 = 1/5$. Independence and symmetry give, for a finite sum,

$$\mathbb{E}X^4 = \frac{1}{5} \sum a_j^4 + \frac{2}{3} \sum_{i < j} a_i^2 a_j^2 = \frac{1}{3} \left(\sum a_j^2 \right)^2 - \frac{2}{15} \sum a_j^4.$$

The infinite series is bounded by $\sum a_j \leq \sqrt{S}/(1-\rho)$, so dominated convergence passes these formulas to the limit. The geometric series for $T(g)$ and theorem 2.1 finish the proof. $\square$

Theorem 3.2 (Quantitative geometric rigidity). *For every $a \in \mathcal{C}_{\rho,S}$,*

$$d_\infty(a, g)^2 \leq S\mathcal{D}_q(\Delta) \leq \frac{S}{2c_q} \Delta = \frac{15}{4c_q S} e_4(a), \quad (3.3)$$

$$|a_j - g_j| \leq \frac{S\mathcal{D}_q(\Delta)}{g_j} \leq \frac{15}{4c_q S g_j} e_4(a). \quad (3.4)$$

The intersection of $\mathcal{C}_{\rho,S}$ with the fourth-moment fibre $m_4(a) = m_4(g)$ is the singleton $\{g\}$.

Proof. For nonnegative x, y , $(\sqrt{x} - \sqrt{y})^2 \leq |x - y|$. Apply this to (2.6) and use theorem 2.2. For the coordinate estimate, use $|a_j - g_j| = |a_j^2 - g_j^2|/(a_j + g_j) \leq Sd/g_j$. If $e_4 = 0$, theorem 2.1 forces $p = b$; nonnegativity then gives $a = g$. The converse follows by substitution. $\square$

Equivalently, the geometric reference uniquely maximizes kurtosis within the fixed-variance cone, because

$$\frac{m_4(a)}{m_2(a)^2} = 3 - \frac{6}{5} \frac{\sum a_j^4}{S^2}.$$

This is a constrained extremal statement, not a generic finite-moment identifiability assertion.

Proposition 3.3 (The same exponent for summable coefficient losses). *Write $p_1 = 1 - q + d$. Then*

$$\|a - g\|_1 \leq \frac{2}{1 - \rho} \left(\frac{p_1}{1 - q} \right)^{1/4} \sqrt{Sd}. \quad (3.5)$$

Consequently every ℓ^p coefficient loss, $1 \leq p \leq \infty$, is $O_{\rho,S}(\sqrt{e_4(a)})$.

Proof. Put $A = \sqrt{S(1 - q)}$, $B = \sqrt{Sp_1}$, and $Z = Sd$. The case $Z = 0$ follows from rigidity. Otherwise, both the envelope and (2.6) give

$$|a_j - g_j| \leq \min\{B\rho^{j-1}, Z/(A\rho^{j-1})\}.$$

Split the geometric sequence at $\rho^{j-1} \approx t$, where $t = \sqrt{Z/(AB)}$. If $t < 1$, the sum on either side of this crossing is at most $\sqrt{ZB/A}/(1 - \rho)$: on the head use the increasing reciprocal geometric bound, and on the tail the decreasing direct bound. If $t \geq 1$, the direct bound alone is at most the same quantity. This proves (3.5). Since $p_1 \leq 1$, theorem 2.2 makes its constant uniform. Finally $\|x\|_p \leq \|x\|_1^{1/p} \|x\|_\infty^{1-1/p}$. $\square$

3.2 An exact support endpoint cannot also move

Theorem 3.4 (Support extremality and rigidity). *For $a \in \mathcal{C}_{\rho,S}$,*

$$\sum_j a_j \leq B_{\rho,S} := \sqrt{\frac{S(1 + \rho)}{1 - \rho}}, \quad (3.6)$$

with equality if and only if $a = g$. The support of μ_a is exactly $[-\sum a_j, \sum a_j]$.

Proof. Set $L = \sum a_j > 0$. Apply theorem 2.1 to a_j/L with ratio parameter ρ , rather than to a_j^2/S with parameter q . It yields $S/L^2 \geq (1 - \rho)/(1 + \rho)$, with equality exactly for a geometric sequence of ratio ρ . The fixed squared sum then determines its scale.

The support containment is immediate from $|U_j| \leq 1$. Each finite uniform sum has its entire Minkowski-sum interval as support. Zero belongs to the support of any tail: choose a sufficiently long finite part near zero and bound the remaining deterministic tail length. Thus every point strictly between $-L$ and L is in the support of the full sum, by using a sufficiently long head whose support contains that point. Closure includes the endpoints. $\square$

At the dyadic boundary this says $\sum a_j \leq 1$. Therefore the simultaneous requirements $a_{j+1} \leq a_j/2$, $\sum a_j^2 = 1/3$, and $\sum a_j = 1$ leave *no nontrivial perturbations at all*. This resolves the exact-support issue at the critical boundary only. It does not resolve the predecessor's exact-support construction problem when slack $\rho > 1/2$ is allowed.

3.3 From moment error to law error

If two probability measures are supported in $[-B, B]$, then

$$|m_4(\mu) - m_4(\nu)| \leq B^4 \text{TV}(\mu, \nu), \quad |m_4(\mu) - m_4(\nu)| \leq 2B^4 d_K(\mu, \nu). \quad (3.7)$$

For the first inequality, x^4/B^4 lies in $[0, 1]$ and the layer-cake representation bounds its expectation difference by total variation. For the second, Stieltjes integration by parts gives $-\int_{-B}^B 4x^3(F_\mu - F_\nu) dx$; the boundary terms vanish when one first integrates on a slightly larger common interval. The integral of $4|x|^3$ is $2B^4$.

For $q = 1/4$, $S = 1/3$, we have $c_q = 3/5$, $B_{\rho, S} = 1$, and

$$m_4(a^0) = \frac{1}{27} - \frac{2}{225} = \frac{19}{675}. \quad (3.8)$$

Substituting in theorem 3.2 and (3.7) proves all upper estimates in theorem 1.2.

For $\rho < 1/2$ the dyadic reference is not in $\mathcal{C}_{\rho, 1/3}$. There is in fact a positive separation, not just a failure of membership:

$$\text{TV}(\mu_a, \mu_0) \geq \eta_\rho, \quad d_K(\mu_a, \mu_0) \geq \frac{1}{2}\eta_\rho, \quad \eta_\rho = \frac{2}{135} \left(\frac{1 - \rho^2}{1 + \rho^2} - \frac{3}{5} \right) > 0. \quad (3.9)$$

Indeed the fourth-power minimum is larger than the dyadic minimum by the quantity displayed, and all supports still lie in $[-1, 1]$.

3.4 When the variance is measured rather than fixed

Suppose $a_{j+1} \leq \rho a_j$, but $S = \sum a_j^2$ is not prescribed. For exact moment measurements with $m_2 > 0$,

$$S = 3m_2, \quad T = \frac{15}{2}(3m_2^2 - m_4), \quad \Delta = \frac{T}{S^2} - c_q. \quad (3.10)$$

Let $g(S_0)$ be a fixed geometric reference. The triangle inequality and theorem 3.2 give the finite certificate

$$d_\infty(a, g(S_0)) \leq \sqrt{S\mathcal{D}_q(\Delta)} + \sqrt{1 - q} |\sqrt{S} - \sqrt{S_0}|. \quad (3.11)$$

Under a common support bound, law convergence to $\mu_{g(S_0)}$ controls both moments. Near $S_0 > 0$, the rational expression in (3.10) is locally Lipschitz, so $|S - S_0| = O(D)$ and $0 \leq \Delta = O(D)$ for $D = \text{TV}$ or d_K . Equation (3.11) therefore retains the exponent $1/2$. With noisy moments, one must propagate measurement intervals; a negative plug-in value of Δ is not itself a valid certificate.

4 Smooth sharpness witnesses inside the original class

4.1 Exact derivative norms for separated supports

Lemma 4.1 (Separated-uniform derivative norms). *Let all a_j be positive, $\sum a_j < \infty$, and assume $a_j \geq \sum_{k>j} a_k$ for every j . Then X_a has a density $f_a \in C_c^\infty(\mathbb{R})$ and*

$$\|f_a^{(r)}\|_\infty = \frac{1}{2^{r+1} \prod_{j=1}^{r+1} a_j} \quad (r = 0, 1, 2, \dots). \quad (4.1)$$

The hypothesis holds whenever $a_{j+1} \leq a_j/2$.

Proof. The characteristic function is $\prod_j \text{sinc}(a_j t)$, with $\text{sinc } u = \sin(u)/u$. Each fixed positive factor contributes a bound $\min(1, 1/(a_j |t|))$. Using arbitrarily many factors gives arbitrarily fast polynomial decay, so Fourier inversion gives a C^∞ density. Its support is compact by absolute summability.

The derivative of the uniform density on $[-a, a]$, in the sense of distributions, is $(\delta_{-a} - \delta_a)/(2a)$. Differentiating each of the first r factors once expresses $f_a^{(r)}$ as a signed sum of translates of the remaining tail density, with coefficient magnitude $1/\prod_{j \leq r} (2a_j)$. These translates have disjoint support interiors. Indeed, if two sign patterns first differ at index i , their centers differ by at least

$$2 \left(a_i - \sum_{i < k \leq r} a_k \right) \geq 2 \sum_{k > r} a_k,$$

which is the diameter of a tail support. Their boundaries cause no additional values because all tail densities are smooth and vanish at their support endpoints.

The density of the tail starting at $r+1$ has supremum exactly $1/(2a_{r+1})$. Its first uniform factor gives this upper bound, and at zero it attains the bound because the rest of the tail is supported in $[-a_{r+1}, a_{r+1}]$. Taking the supremum of the disjoint translates proves (4.1). Finally $\sum_{k > j} a_k \leq a_j \sum_{k \geq 1} 2^{-k} = a_j$ under the ratio bound. $\square$

For the dyadic reference this gives

$$\|f_0^{(r)}\|_\infty = 2^{r(r+1)/2}. \quad (4.2)$$

This special-case identity is also recorded in the repository's formal crosswalk [1]. We have supplied an independent proof of the more general separated-support formula rather than assuming that crosswalk as a formal verification of this manuscript.

Definition 4.2 (The predecessor's smooth class at the boundary). Let $\mathcal{K}$ consist of decreasing nonnegative spectra with $\sum a_j \leq 3$, variance $1/9$, and a density $f_a \in C_c^\infty(\mathbb{R})$ satisfying

$$\|f_a^{(r)}\|_\infty \leq 2^{r+1} \|f_0^{(r)}\|_\infty \quad (r \geq 0). \quad (4.3)$$

This is the class used in the predecessor [1]. Define $\mathcal{K}_{1/2} = \mathcal{K} \cap \mathcal{C}_*$. The new notation extends its ratio-restricted classes to the previously unresolved endpoint.

4.2 Delete one factor and restore the variance

For $n \geq 3$, put

$$h_n = 2^{-n}, \quad c_n = (1 - 3h_n^2)^{-1/2}, \quad a_j^{(n)} = \begin{cases} c_n 2^{-j}, & j < n, \\ c_n 2^{-(j+1)}, & j \geq n. \end{cases} \quad (4.4)$$

This is the spectrum obtained by deleting the n th dyadic uniform factor, reindexing, and rescaling all remaining factors to restore the variance. Unlike a truncation, it keeps infinitely many positive factors and therefore keeps a smooth density.

Proposition 4.3 (Exact properties of the deleted-factor family). *For every $n \geq 3$, $a^{(n)} \in \mathcal{K}_{1/2}$. Writing $h = h_n$ and $c = c_n$,*

$$\sum_j a_j^{(n)} = c(1 - h) \leq 1, \quad (4.5)$$

$$d_\infty(a^{(n)}, a^0) = \max\{(c - 1)/2, h(1 - c/2)\} \geq h/3, \quad (4.6)$$

$$e_4(a^{(n)}) = \frac{4}{75} \frac{h^2(1 - 4h^2)}{(1 - 3h^2)^2}, \quad (4.7)$$

$$\text{TV}(\mu_{a^{(n)}}, \mu_0) \leq 4h^2. \quad (4.8)$$

In particular $d_\infty(a^{(n)}, a^0)/h \rightarrow 1/2$ and $e_4(a^{(n)})/h^2 \rightarrow 4/75$.

Proof. All ratios are $1/2$ except at $n - 1$, where the ratio is $1/4$. The squared sum is $c^2(1/3 - h^2) = 1/3$. The support estimate follows from $(1 - h)^2 \leq 1 - 3h^2$, valid for $h \leq 1/2$.

Before index n , the largest coefficient change is $(c - 1)/2$. From index n onward, it is $h(1 - c/2)$. Since $n \geq 3$ gives $c \leq 4/3$, this proves (4.6). The fourth-power sum is $c^4(1/15 - h^4)$, and direct subtraction yields

$$\sum_j (a_j^{(n)})^4 - \frac{1}{15} = \frac{2}{5} \frac{h^2(1 - 4h^2)}{(1 - 3h^2)^2}.$$

Equation (3.1) proves (4.7).

For every derivative order r , the ratio of the first $r + 1$ coefficient products to the corresponding dyadic product is $c^{r+1}2^{-\max(0, r-n+2)}$. By theorem 4.1,

$$\frac{\|f_{a^{(n)}}^{(r)}\|_\infty}{\|f_0^{(r)}\|_\infty} = c^{-(r+1)}2^{\max(0, r-n+2)} \leq 2^{r+1}. \quad (4.9)$$

This checks all the infinitely many inequalities in (4.3), not merely a finite set of derivative orders.

For the total variation estimate, let v_n be the density of the dyadic sum with the n th factor deleted. The original density is $f_0 = v_n * u_h$, where u_h is the uniform density on $[-h, h]$. The first two dyadic factors are still present. Distributional differentiation of those factors, followed by convolution, gives

$$\|v_n'\|_1 \leq 2, \quad \|v_n''\|_1 \leq 8. \quad (4.10)$$

The centered Taylor remainder gives

$$\|v_n * u_h - v_n\|_1 \leq \frac{h^2}{6} \|v_n''\|_1, \quad \text{TV}(v_n * u_h, v_n) \leq \frac{2}{3} h^2.$$

For the dilation operator $D_c v(x) = c^{-1}v(x/c)$, differentiating with respect to $t = \log c$ and changing variables yields

$$\|D_c v - v\|_1 \leq |\log c| \|v + xv'\|_1.$$

Since v_n is supported in $[-1, 1]$, (4.10) gives $\|v_n + xv_n'\|_1 \leq 3$. Thus the additional total variation from rescaling is at most $3 \log c/2$. Finally

$$\frac{2}{3} h^2 + \frac{3}{2} \log c \leq h^2 \left(\frac{2}{3} + \frac{9}{4(1 - 3h^2)} \right) < 4h^2 \quad (h \leq 1/8),$$

using $-\log(1 - x) \leq x/(1 - x)$. This proves (4.8). The two limits follow from the explicit formulas. $\square$

Definition 4.4 (Anchored modulus). For a class $\mathcal{A}$ containing a^0 and a law distance D , define

$$\omega_{\mathcal{A},D}(\varepsilon) = \sup\{d_\infty(a, a^0) : a \in \mathcal{A}, D(\mu_a, \mu_0) \leq \varepsilon\}. \quad (4.11)$$

Theorem 4.5 (Sharp boundary modulus in the exact smooth class). For $\mathcal{A} = \mathcal{C}_*$ or $\mathcal{A} = \mathcal{K}_{1/2}$, and all sufficiently small $\varepsilon > 0$,

$$\frac{1}{12}\sqrt{\varepsilon} \leq \omega_{\mathcal{A},\text{TV}}(\varepsilon) \leq \sqrt{\frac{75}{4}}\sqrt{\varepsilon}, \quad (4.12)$$

$$\frac{1}{12}\sqrt{\varepsilon} \leq \omega_{\mathcal{A},d_K}(\varepsilon) \leq \sqrt{\frac{75}{2}}\sqrt{\varepsilon}. \quad (4.13)$$

No exponent strictly greater than $1/2$ is valid in either class. The same sharp exponent holds for every ℓ^p coefficient loss, $1 \leq p \leq \infty$.

Proof. The upper estimates were proved in section 3. For the lower bound, choose the smallest $n \geq 3$ with $4h_n^2 \leq \varepsilon$. For sufficiently small ε , minimality gives $16h_n^2 > \varepsilon$, hence $h_n > \sqrt{\varepsilon}/4$. By theorem 4.3, the law distance is at most ε and the spectral gap is greater than $\sqrt{\varepsilon}/12$. Kolmogorov distance is no larger than total variation, so the same witnesses work there. The ℓ^p upper bounds follow from theorem 3.3 and (3.7); their lower bounds follow from $\|a - a^0\|_p \geq d_\infty(a, a^0)$. $\square$

The constants in (4.12)–(4.13) are not asserted to be optimal. Sharpness here means the exact exponent and a matching rate for every sufficiently small error level, not merely along a subsequence.

5 A Hellinger estimate that sees the support boundary

The total variation bound $O(h^2)$ alone gives only an $\Omega(h^{-2})$ testing lower bound by the elementary product-TV inequality. To obtain the correct h^{-4} scale, we need squared Hellinger distance of order h^4 . Because the laws have compact supports, a careless division by a density outside its support would be invalid. The following lemmas explicitly handle that boundary.

We use the convention

$$H^2(f, g) = \int_{\mathbb{R}} (\sqrt{f} - \sqrt{g})^2 dx. \quad (5.1)$$

Then $0 \leq H^2 \leq 2$ and $\text{TV}(f, g) \leq H(f, g)$. Squared Hellinger distance is jointly convex under common mixtures: Cauchy–Schwarz gives

$$\sqrt{\left(\int f_z d\nu(z)\right)\left(\int g_z d\nu(z)\right)} \geq \int \sqrt{f_z g_z} d\nu(z).$$

Integration proves the claim, and convolution contraction is its special case for common mixtures of translations.

Lemma 5.1 (Centered smoothing costs fourth order in squared Hellinger). *Let v be the density of a sum of $m \geq 5$ independent uniforms with fixed positive half-lengths. There exist $C, h_0 > 0$ such that, for every centered random variable Z_h supported in $[-h, h]$, with $0 < h < h_0$,*

$$H^2(v, v * \text{Law}(Z_h)) \leq Ch^4. \quad (5.2)$$

The constant is independent of the distribution of Z_h .

Proof. Write B for the sum of the half-lengths, and $v_h = v * \text{Law}(Z_h)$. The convolution density v is a C^{m-2} piecewise polynomial, positive on $(-B, B)$. For sufficiently small $s > 0$ it has the exact endpoint form

$$v(-B + s) = v(B - s) = C_0 s^{m-1}, \quad C_0 = \frac{1}{(m-1)! \prod_{i=1}^m (2\alpha_i)}, \quad (5.3)$$

where α_i are the half-lengths. This follows by convolving constant densities on $[0, 2\alpha_i]$: before reaching the smallest upper endpoint, the convolution is the elementary simplex-volume polynomial.

On the interior $I_h = [-B + 2h, B - 2h]$, Taylor's theorem and $\mathbb{E}Z_h = 0$ give

$$|v_h(x) - v(x)| \leq \frac{h^2}{2} \sup_{|u-x| \leq h} |v''(u)|. \quad (5.4)$$

Also $(\sqrt{v_h} - \sqrt{v})^2 \leq (v_h - v)^2/v$ there. If s is the distance to a nearby endpoint and $s \geq 2h$, the endpoint formula gives

$$\frac{\sup_{|u-x| \leq h} |v''(u)|^2}{v(x)} \leq C_1 s^{m-5}.$$

This is uniformly integrable because $m \geq 5$. On a fixed central compact interval, positivity of v and boundedness of v'' give the same uniform integral bound. Thus the Hellinger integral over I_h is at most $C_2 h^4$.

Outside I_h , both densities can be nonzero only within distance $3h$ of the old endpoints when their defining convolution arguments are considered. Equation (5.3) bounds the total mass of either density in these boundary and exterior strips by $C_3 h^m$. Since $(\sqrt{v_h} - \sqrt{v})^2 \leq v_h + v$, their contribution is $O(h^m) = O(h^4)$. This also covers the part where $v = 0$ and v_h need not vanish. Combining the two regions proves the result. $\square$

Lemma 5.2 (Uniform Hellinger control of small dilations). *For the density v in theorem 5.1 and fixed $R < \infty$, there is $A_R < \infty$ such that every probability measure ν supported in $[-R, R]$ and every $c > 0$ satisfy*

$$H(D_c(v * \nu), v * \nu) \leq A_R |\log c|. \quad (5.5)$$

Proof. First fix a translate $z \in [-R, R]$, and put $c = e^t$. After translation by $-z$, comparison of $D_c(v(\cdot - z))$ with $v(\cdot - z)$ becomes comparison of v with the density of $e^t Y + (e^t - 1)z$, where Y has density v . Its square root is

$$s_t(x) = e^{-t/2} \sqrt{v(e^{-t}(x + z) - z)}.$$

Differentiation and the substitution $u = e^{-t}(x + z) - z$ give

$$\|\partial_t s_t\|_2 = \frac{1}{2} \left(\int_{-B}^B \frac{(v(u) + (u + z)v'(u))^2}{v(u)} du \right)^{1/2}. \quad (5.6)$$

The integral is finite uniformly in $|z| \leq R$: near an endpoint the potentially singular term $v'(u)^2/v(u)$ is $O(s^{m-3})$, and on central compact intervals v is bounded below. There are no distributional boundary terms for the square root. Indeed (5.3) shows that $\sqrt{v}$, extended by zero, lies in $H^1(\mathbb{R})$ and vanishes at the endpoints. The displayed differentiation is justified first for smooth compactly supported approximations to $\sqrt{v}$ and then in L^2 , using convergence of the function and its derivative; the supports are uniformly bounded.

The right side of (5.6) is independent of t and bounded by some A_R . The fundamental theorem of calculus in L^2 therefore gives $\|s_t - s_0\|_2 \leq A_R |t|$. Finally mix over the same $z \sim \nu$ and use joint convexity of H^2 . $\square$

Theorem 5.3 (Hellinger order of the deleted-factor witnesses). *There are constants $0 < c_H \leq C_H < \infty$ such that, for every $n \geq 6$,*

$$c_H h_n^4 \leq H^2(\mu_{a^{(n)}}, \mu_0) \leq C_H h_n^4. \quad (5.7)$$

Proof. Let v be the density of the first five dyadic uniforms. The density v_n of the sum with the n th factor deleted has the form $v * \nu_n$, with ν_n supported in a fixed bounded interval. By convolution contraction and theorem 5.1,

$$H(v_n, v_n * u_{h_n}) \leq H(v, v * u_{h_n}) \leq C h_n^2.$$

By theorem 5.2, $H(D_{c_n} v_n, v_n) \leq A \log c_n = O(h_n^2)$. Since $v_n * u_{h_n} = f_0$, the triangle inequality proves the upper bound for all sufficiently large n . Increasing the constant covers the finitely many remaining indices $n \geq 6$.

For the lower bound, (3.7) and $\text{TV} \leq H$ imply $H^2 \geq e_4(a^{(n)})^2$. Formula (4.7) has a positive ratio to h_n^2 tending to $4/75$, so it is bounded below by a positive constant times h_n^2 on all $n \geq 6$. Squaring finishes the proof. $\square$

The proof only needs five unchanged positive uniform factors, centered small noise, and a bounded remaining support. This is why it also extends to nongeometric spectra in section 8. It does not require positivity of a compactly supported density on the entire real line, finite chi-square distance across unequal supports, or an unproved pointwise likelihood expansion.

6 Quartic sample complexity and honest confidence sets

6.1 A self-contained concentration input

For independent $Z_i \in [0, 1]$ with common mean m , the standard Hoeffding bound [4] is

$$\mathbb{P}\{\bar{Z}_N - m \geq t\} \leq e^{-2Nt^2}, \quad \mathbb{P}\{m - \bar{Z}_N \geq t\} \leq e^{-2Nt^2}. \quad (6.1)$$

For completeness, if $\psi(\lambda) = \log \mathbb{E} e^{\lambda(Z - \mathbb{E}Z)}$, then $\psi(0) = \psi'(0) = 0$ and $\psi''(\lambda)$ is a variance under an exponentially tilted law on $[0, 1]$. That variance is at most $1/4$, because $\text{Var } Z \leq \mathbb{E}(Z - 1/2)^2 \leq 1/4$. Hence $\psi(\lambda) \leq \lambda^2/8$. Independence, Markov's inequality, and optimization at $\lambda = 4t$ prove the first bound for the average; apply it to $1 - Z_i$ for the second. This is a proof of the standard input, not a new concentration inequality.

6.2 The testing theorem

For $\mathcal{A} = \mathcal{C}_*$ or $\mathcal{K}_{1/2}$, let $N_{\mathcal{A}}(\delta)$ be the smallest positive integer N admitting a possibly randomized test ψ with

$$\mathbb{E}_{\mu_0^{\otimes N}} \psi \leq \frac{1}{4}, \quad \sup_{\substack{a \in \mathcal{A} \\ d_{\infty}(a, a^0) \geq \delta}} \mathbb{E}_{\mu_a^{\otimes N}} (1 - \psi) \leq \frac{1}{4}. \quad (6.2)$$

Theorem 6.1 (Sharp quartic testing complexity). *For either class, $N_{\mathcal{A}}(\delta) = \Theta(\delta^{-4})$ as $\delta \downarrow 0$. A concrete upper bound is*

$$N_{\mathcal{A}}(\delta) \leq \left\lceil \frac{5625 \log 4}{8\delta^4} \right\rceil. \quad (6.3)$$

Proof. Given observations $X_1, \dots, X_N$, set $\bar{Z}_N = N^{-1} \sum_{i=1}^N X_i^4$ and reject the null if

$$\frac{19}{675} - \bar{Z}_N > \frac{2\delta^2}{75}. \quad (6.4)$$

Every X_i lies in $[-1, 1]$, so $Z_i = X_i^4 \in [0, 1]$. Under the null, (6.1) bounds rejection probability by $\exp(-8N\delta^4/5625)$. Any alternative in (6.2) has $e_4(a) \geq 4\delta^2/75$ by (1.4). Acceptance then requires an upward empirical moment error at least $2\delta^2/75$, with the same probability bound. This proves (6.3).

For the lower bound, choose n so that $3\delta \leq h_n < 6\delta$; for small δ we have $n \geq 6$. By theorem 4.3, this is an alternative in $\mathcal{K}_{1/2}$ at spectral distance at least δ . For any probability laws P, Q , multiplicativity of Hellinger affinity gives

$$H^2(P^{\otimes N}, Q^{\otimes N}) = 2 \left[1 - \left(1 - \frac{1}{2} H^2(P, Q) \right)^N \right] \leq N H^2(P, Q). \quad (6.5)$$

Combining theorem 5.3, (6.5), and $\text{TV} \leq H$ yields

$$\text{TV}(\mu_{a(n)}^{\otimes N}, \mu_0^{\otimes N}) \leq \sqrt{N C_H h_n^4} \leq \sqrt{N C_H 6^4 \delta^4}.$$

For $N \leq (16C_H 6^4)^{-1} \delta^{-4}$ this is at most $1/4$. The two errors of any test at these two laws sum to at least one minus this total variation, hence at least $3/4$. They cannot both be at most $1/4$. This proves the lower order for $\mathcal{K}_{1/2}$ and therefore for $\mathcal{C}_*$ as well. $\square$

The constant in the lower bound is not numerically optimized. Its finiteness is established analytically by the explicit spline endpoint and dilation-score integrals. No numerical Hellinger integration is being used as a proof.

6.3 Confidence sets and an observable radius

For $0 < \alpha < 1$, set

$$t_N = \sqrt{\frac{\log(2/\alpha)}{2N}}, \quad \mathcal{F}_N = \{a \in \mathcal{C}_* : |m_4(a) - \bar{Z}_N| \leq t_N\}. \quad (6.6)$$

By the two tails of (6.1), $\mathbb{P}_a\{a \in \mathcal{F}_N\} \geq 1 - \alpha$ uniformly over $\mathcal{C}_*$. At the reference, on an event of probability at least $1 - \alpha$, every candidate in $\mathcal{F}_N$ satisfies $e_4(a) \leq 2t_N$. Hence

$$\sup_{a \in \mathcal{F}_N} d_\infty(a, a^0) \leq \sqrt{\frac{75}{2}} t_N, \quad (6.7)$$

$$\sup_{a \in \mathcal{F}_N} d_r(a, a^0) \leq \frac{75}{2} 2^r t_N. \quad (6.8)$$

On that event $a^0 \in \mathcal{F}_N$, so no convention about the supremum of an empty confidence set is needed. This establishes honest coverage and reference-point contraction, not a finite representation of the entire infinite-dimensional set. The scalar radius and test themselves are directly computable from a fourth-moment average.

7 Sharp local conditioning of a growing prefix

For $D = \text{TV}$ or d_K and $\mathcal{A} = \mathcal{C}_*$ or $\mathcal{K}_{1/2}$, define

$$\Lambda_r^D(\mathcal{A}) = \lim_{\varepsilon \downarrow 0} \sup_{\substack{a \in \mathcal{A} \\ 0 < D(\mu_a, \mu_0) \leq \varepsilon}} \frac{d_r(a, a^0)}{D(\mu_a, \mu_0)}. \quad (7.1)$$

The limit exists as a decreasing limit of suprema, allowing infinity in principle. The earlier upper bounds make it finite here.

Theorem 7.1 (The prefix constant grows exactly geometrically). *For every $r \geq 6$ and either class,*

$$\frac{2^r}{30} \leq \Lambda_r^{\text{TV}}(\mathcal{A}) \leq \frac{75}{4} 2^r, \quad (7.2)$$

$$\frac{2^r}{30} \leq \Lambda_r^{d_K}(\mathcal{A}) \leq \frac{75}{2} 2^r. \quad (7.3)$$

Proof. The upper estimates follow directly from (1.5) and (3.7). For the lower estimates set $n = r$, $h = 2^{-n}$, and, for $1/2 \leq \theta < 1$, put

$$\lambda = 1 - \theta^2, \quad c = (1 - 4\lambda h^2)^{-1/2}, \quad a_j(\theta) = \begin{cases} c2^{-j}, & j < n, \\ c\theta 2^{-j}, & j \geq n. \end{cases} \quad (7.4)$$

The dyadic tail starting at n has squared coefficient sum $4h^2/3$. Thus the chosen c restores squared sum $1/3$. The only altered ratio is at $n - 1$, where it is $\theta/2 \leq 1/2$. For every derivative order k , theorem 4.1 gives the norm ratio

$$c^{-(k+1)} \theta^{-\max(0, k-n+2)} \leq 2^{k+1}.$$

Together with theorem 3.4, this proves $a(\theta) \in \mathcal{K}_{1/2}$.

At coordinate n , $c\theta < 1$, and

$$1 - c\theta = \frac{\lambda(1 - 4h^2)}{(1 - 4\lambda h^2)(1 + c\theta)} \geq \frac{1 - \theta}{3}. \quad (7.5)$$

Here the denominator is at most 2, $\lambda \geq 1 - \theta$, and $h \leq 1/8$. Consequently $d_n(a(\theta), a^0) \geq (1 - \theta)h/3$.

We next show

$$\text{TV}(\mu_{a(\theta)}, \mu_0) \leq 10(1 - \theta)h^2. \quad (7.6)$$

Let w be the density of the head before n , and let T be the independent dyadic random tail starting at n . Then $\mathbb{E}T = 0$ and $\mathbb{E}T^2 = 4h^2/9$. For $s \in [\theta, 1]$, differentiating $\mathbb{E}w(x - sT)$ and subtracting $\mathbb{E}[Tw'(x)] = 0$ yields

$$\left\| \frac{d}{ds} \mathbb{E}w(\cdot - sT) \right\|_1 \leq s\mathbb{E}T^2 \|w''\|_1.$$

Integration gives an L^1 difference at most $\lambda\mathbb{E}T^2 \|w''\|_1/2$. Since the head contains the first five factors, $\|w''\|_1 \leq 8$. The resulting TV bound is $8\lambda h^2/9$. The law before rescaling retains a first uniform factor of half-length $1/2$, has derivative L^1 norm at most 2, and is supported in $[-1, 1]$. Its rescaling costs TV at most

$$\frac{3}{2} \log c \leq \frac{3\lambda h^2}{1 - 4\lambda h^2} \leq 4\lambda h^2.$$

The total is at most $5\lambda h^2 \leq 10(1 - \theta)h^2$, proving (7.6).

For $\theta < 1$ the spectrum is not geometric, so moment rigidity ensures that both law distances are positive. As $\theta \uparrow 1$, they tend to zero by (7.6). Dividing (7.5) times h by (7.6) gives a limiting ratio at least $1/(30h) = 2^r/30$. The same lower bound applies to $d_K \leq \text{TV}$. Since these witnesses are in $\mathcal{K}_{1/2}$, it applies to both classes. $\square$

This conditioning law is substantially more favorable than the $\exp(O(r^2))$ upper constants in the predecessor's prefix theorem with an unrestricted infinite tail [1]. It is a comparison between *different constraint classes*, not a correction to that theorem. The exact critical constraint supplies a new order structure that the larger class does not possess.

8 Nongeometric reference spectra and relative-ratio cones

The geometric zero lattice is not needed for the rigidity argument. The appropriate general structure is monotonicity of the coefficient ratio to a fixed reference, which translates into a monotone likelihood ratio between two probability sequences.

Let $b = (b_j)_{j \geq 1}$ be positive with

$$\sum_j b_j^2 = S > 0, \quad 0 < r_- \leq \frac{b_{j+1}}{b_j} \leq r_+ < 1. \quad (8.1)$$

Define the relative-ratio cone

$$\mathcal{C}_b = \left\{ a_j \geq 0 : \sum_j a_j^2 = S, \quad \frac{a_{j+1}}{b_{j+1}} \leq \frac{a_j}{b_j} \quad (j \geq 1) \right\}. \quad (8.2)$$

In particular $a_{j+1} \leq r_+ a_j$, so there is a common support bound $\sum a_j \leq \sqrt{S}/(1 - r_+)$. Write

$$w_j = b_j^2/S, \quad p_j = a_j^2/S, \quad \Delta_b = \sum_j p_j^2 - \sum_j w_j^2.$$

Theorem 8.1 (Relative-ratio moment rigidity). *For every $a \in \mathcal{C}_b$,*

$$\Delta_b \geq 0, \quad d_\infty(a, b)^2 \leq \frac{S \Delta_b}{2w_1(w_1 - w_2)} = \frac{15 [m_4(b) - m_4(a)]}{4S w_1(w_1 - w_2)}. \quad (8.3)$$

Equality of the second and fourth moments with those of b forces $a = b$. For this upper conclusion, the lower ratio bound $r_- > 0$ is unnecessary; it suffices that the positive reference has $w_1 > w_2$, is decreasing, and is summable, with a common support bound for the metric consequence.

Proof. The sequence $p_j/w_j = (a_j/b_j)^2$ is nonincreasing, with w -weighted average one. Its weighted average on any leading block is at least its weighted average on the complementary tail, and therefore is at least one. Thus $D_k = \sum_{j \leq k} (p_j - w_j) \geq 0$. Summation by parts, exactly as in theorem 2.1, gives

$$\Delta_b = \sum_j (p_j - w_j)^2 + 2 \sum_{k \geq 1} (w_k - w_{k+1}) D_k. \quad (8.4)$$

The terms are nonnegative and the sums converge. Set $d = p_1 - w_1 = D_1$. Then $\Delta_b \geq 2(w_1 - w_2)d$.

Since $p_j/w_j \leq p_1/w_1 = 1 + d/w_1$, the sum of the positive differences $p_j - w_j$ is at most d/w_1 . Its value equals the absolute sum of the negative differences because both sequences have total mass one. Hence $|p_j - w_j| \leq d/w_1$ at every coordinate. Taking square roots as in theorem 3.2 proves the first bound in (8.3). The moment formula (3.1) proves the second. If the moment deficit is zero, the first term of (8.4) vanishes and $p = w$, hence $a = b$. $\square$

Theorem 8.2 (Sharp recovery and testing for a fixed ratio profile). *Under (8.1), the modulus anchored at b over $\mathcal{C}_b$ is $\Theta(\sqrt{\varepsilon})$ for either TV or Kolmogorov distance. Testing b against alternatives $a \in \mathcal{C}_b$ with $d_\infty(a, b) \geq \delta$ has sample complexity $\Theta(\delta^{-4})$ at fixed errors $1/4$. The lower witnesses have infinitely many positive coefficients and compactly supported C^∞ densities.*

Proof. The upper modulus follows from theorem 8.1 and (3.7) with the common support bound just stated. For lower witnesses, fix $\theta = 1/2$, and let

$$\tau_n = \sum_{j \geq n} b_j^2, \quad c_n = \left(1 - (1 - \theta^2)\tau_n/S\right)^{-1/2}, \quad a_j^{[n]} = \begin{cases} c_n b_j, & j < n, \\ c_n \theta b_j, & j \geq n. \end{cases} \quad (8.5)$$

These spectra belong to $\mathcal{C}_b$, and

$$b_n^2 \leq \tau_n \leq \frac{b_n^2}{1 - r_+^2}, \quad c_n - 1 = O_b(b_n^2), \quad d_\infty(a^{[n]}, b) \geq b_n/4$$

for all sufficiently large n . The last inequality follows at coordinate n from $c_n \theta \rightarrow 1/2$.

Use the first five reference uniforms as the fixed convolution density v in theorem 5.1. The remaining head before n supplies a common convolution factor. The centered tail starting at n is supported in an interval of radius at most $b_n/(1 - r_+)$, and its contracted version has the same type of bound. Comparing each with the unchanged head by theorem 5.1 and then using the triangle inequality gives Hellinger distance $O_b(b_n^2)$ between the unscaled laws. Rescaling adds $O_b(\log c_n) = O_b(b_n^2)$ by theorem 5.2, whose residual support bound is uniform. Therefore

$$H^2(\mu_{a^{[n]}}, \mu_b) = O_b(b_n^4), \quad \text{TV}(\mu_{a^{[n]}}, \mu_b) = O_b(b_n^2). \quad (8.6)$$

Positivity of every coefficient gives smoothness by the Fourier-decay argument in theorem 4.1; separated translates are not needed for this part.

The lower ratio bound $b_{n+1} \geq r_- b_n$ lets us choose b_n within a fixed multiplicative factor of any sufficiently small target scale. Thus (8.6) and the spectral gap imply a lower modulus of order $\sqrt{\varepsilon}$ for every sufficiently small ε , not just selected errors. For testing, the fourth-moment deficit in (8.3) is at least a fixed multiple of δ^2 , so the bounded-variable argument in theorem 6.1 gives an $O_b(\delta^{-4})$ upper bound. Choose n with $4\delta \leq b_n < 4\delta/r_-$. Equations (8.6) and (6.5) give the matching lower bound just as before. $\square$

Every geometric reference in (2.1) satisfies (8.1) with $r_- = r_+ = \rho$, and its relative-ratio cone is exactly $\mathcal{C}_{\rho, S}$. Thus the sharp exponent and quartic testing law hold for every geometric reference, not only the dyadic one. For those geometric references, theorem 3.3 supplies the matching upper bounds for all ℓ^p losses; the same witnesses supply the lower bounds.

Without a positive lower ratio bound, theorem 8.1 still gives its upper estimate, and (8.6) still obstructs every better Hölder exponent along a sequence. What may fail is interpolation between widely separated scales. No all-error matching modulus is claimed for that more general case.

9 What has been resolved, and what has not

Question or regime	Conclusion and precise scope
Exact dyadic ratio 1/2	Sharp square-root anchored modulus, including the predecessor's exact smooth class; theorem 4.5.
Fixed variance and two moments	The dyadic and geometric moment fibres at the reference are singletons inside their critical cones; theorem 3.2.
Exact maximal support	With the critical ratio and fixed variance, fixing the reference support endpoint forces the reference itself; theorem 3.4.
Leading r coefficients	Local conditioning is $\Theta(2^r)$ in the critical dyadic cone, not uniformly over unrestricted tails; theorem 7.1.
Statistical complexity	Anchored testing is $\Theta(\delta^{-4})$; honest reference contraction is $O(N^{-1/4})$; theorem 6.1.
Looser dyadic ratio bound	The predecessor's fixed-slack stretched-exponential result is cited, not reproved or newly formalized here.
Arbitrary two-point inverse	Not settled. The moment extremizer is a fixed anchor; the proofs do not compare two free spectra.

The apparent abrupt transition at 1/2 has a precise cause. At the exact boundary, normalized squared coefficients are forced to majorize the reference sequence. Their fourth-power deficit cannot be canceled. With slack, that one-sided relation to the dyadic reference disappears, permitting the moment-matching perturbations investigated by the predecessor. This explanation is supported by (2.3), rather than by numerical conditioning experiments.

The results are not claimed as a resolution of a famous general inverse problem. They resolve a specific, explicitly stated research frontier in the repository and give additional quantitative consequences with complete proofs. Their relevance is the contrast between severe infinite-dimensional instability and a two-moment certificate under an exact structural constraint.

10 Verification, provenance, and formalization route

10.1 Exact computations and their limits

The accompanying standard-library Python program `code/verify.py` uses exact rational arithmetic for finite regression tests and 65-digit `Decimal` arithmetic only for illustrative radical evaluations. The recorded run used Python 3.13.5, deterministic seed 20260930, and passed **29,776 exact assertions**. The detailed counts are in `results/verification.json`. They include the infinite-tail energy identity evaluated symbolically for finite padded sequences, the sharp head inequality, squared-coefficient bounds, exact fourth-moment formulas, derivative-class inequalities for tested orders, tail-deformation estimates, and nongeometric finite-profile identities.

These are finite regression checks. In particular, checking derivative orders $0 \leq r \leq 100$ is *not* evidence for all orders by itself; the universal proof is (4.9). The Hellinger lemmas and infinite limiting assertions are proved in the text, not certified by numerical tests. The program performs no empirical integration of TV or Hellinger distance.

Run `python code/verify.py` from the package root, or by absolute path from any directory. By default the outputs go to `results-rerun/`, so the recorded results are preserved. The option `--output-dir PATH` selects another destination. The TeX source contains its tables directly and does not require Python or network access to compile. Run `latexmk -pdf article.tex`, or run `pdflatex article.tex` repeatedly until cross-references stabilize. No shell escape is required.

n	Spectral gap	Fourth-moment deficit	TV upper bound $4h_n^2$
3	$6.0981560 \cdot 10^{-2}$	$8.5998388 \cdot 10^{-4}$	$6.2500000 \cdot 10^{-2}$
4	$3.1065269 \cdot 10^{-2}$	$2.0997047 \cdot 10^{-4}$	$1.5625000 \cdot 10^{-2}$
6	$7.8096374 \cdot 10^{-3}$	$1.3027193 \cdot 10^{-5}$	$9.7656250 \cdot 10^{-4}$
8	$1.9530803 \cdot 10^{-3}$	$8.1382692 \cdot 10^{-7}$	$6.1035156 \cdot 10^{-5}$
12	$1.2207030 \cdot 10^{-4}$	$3.1789148 \cdot 10^{-9}$	$2.3841858 \cdot 10^{-7}$

Table 1: Rounded evaluations of theorem 4.3. The last column is a proved upper bound, not the actual TV distance. Decimal rounding is not outward-directed interval certification.

10.2 Pinned repository source and question mapping

The source inspection used the repository snapshot

```
f4457a1d19701d32e8f324240c497653d235b53e
```

and the article at

```
Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/
drafts/inverse-and-sampling/Anchored_Dyadic_Recovery_Uniform_Spectrum/
article.tex.
```

Its content blob was

```
50cc46f4084386b48378cbc9c1d942c66ce64741.
```

The opening definitions, the smooth-class and derivative sections, the statistical section, and the further-research questions were read through the GitHub connector. The exact target is the question titled *The sharp separation boundary* $\rho = 1/2$, third in the final question list. The immediately following exact-support question is addressed here only at the critical boundary. The earlier article’s general identifiability, Fourier zero-count arguments, and numerical bounds are not assumptions in our new critical-boundary proofs.

The source’s editorial crosswalk reports the formal derivative-norm theorem `Fabius.isGREATEST_ABS_ITERATEDDERIV_RVACHEVUP` in `GlobalBounds.lean` and the variance theorem `Fabius.integral_sq_mul_rvachev_eq_one_ninth` in `DyadicSpecializations.lean`, under `Analysis/FabiusFunction/Lean/FabiusFunction/`. These names are recorded as repository provenance. No Lean build was run for this article, and no claim is made that the newly stated stability, Hellinger, or statistical theorems are already formalized.

The literature check was targeted: Arias de Reyna for the compact-support uniform-sum function, Billey–Swanson for generalized uniform-sum identifiability, Hoeffding for concentration, and repository search for related inverse and lacunarity reports. This is not an exhaustive novelty audit. No repository files were modified.

10.3 Claim-level dependency audit

Conclusion	Mathematical dependencies
Two-moment certificate	Tail domination, summation by parts, the sharp head inequality, envelope slacks, and the elementary fourth-moment formula.
Exact support rigidity	The same majorization inequality applied to normalized coefficients rather than their squares.
Smooth lower witnesses	Exact variance rescaling, separated translates of uniform derivatives, and an L^1 centered Taylor remainder.
Fourth-order Hellinger	Fixed finite uniform convolution, endpoint simplex polynomials, separate boundary integration, and a square-root dilation score.
Sharp testing order	Fourth-moment exposure, bounded-variable concentration, the explicit witnesses, and product Hellinger affinity.
Sharp prefix constants	Global coordinate certificate and a continuous variance-preserving contraction of an entire tail.
Nongeometric extension	Monotone relative ratios, a general weighted summation-by-parts identity, and the same smoothing mechanism.

A practical formalization route begins with finite partial-sum identities and nonnegative geometric tails. One can then formalize the probability-sequence energy identity, the quadratic head estimate, and the envelope-slack argument without any Fabius-specific analysis. The fourth-moment identity is a separate finite-independence lemma followed by dominated convergence. These modules already give the main upper certificate and rigidity theorem.

The analytic lower-bound modules should remain separate: convolution smoothness; exact derivative norms from disjoint support interiors; L^1 smoothing and dilation; then the endpoint-sensitive Hellinger lemmas. The latter require an explicit zero-extension and Sobolev differentiation interface, rather than a blanket differentiability assertion for likelihood ratios. Finally, the testing theorem is assembled from deterministic moment exposure and product-measure inequalities. This is a dependency plan, not a claim about the availability of every needed theorem in a current mathlib API.

11 Further research questions

Research question 1 (Optimal constants and possible log-periodic structure). Do the limits $\omega_{C_*,D}(\varepsilon)/\sqrt{\varepsilon}$ exist for $D = \text{TV}$ and d_K ? The present constants are deliberately nonoptimal. The deleted-factor scales are geometric, while continuous tail contractions provide an extra parameter. Determine whether optimal combinations eliminate log-periodic oscillation, and whether the explicit moment constant $75/4$ in the certificate can itself be improved. Sharpness of the exponent does not answer either constant question.

Research question 2 (A joint small-slack crossover). Let the allowed dyadic ratio be $1/2 + \eta$ with $\eta \downarrow 0$ as the law error $\varepsilon \downarrow 0$. Determine the two-parameter inverse modulus. The critical square-root theorem and the predecessor's fixed-positive-slack stretched-exponential theorem concern different orders of limits. A quantitative measure of the negative parts of the cumulative discrepancies in (2.3) may interpolate between the regimes.

Research question 3 (Pairwise recovery within the critical cone). What is the optimal modulus between two freely varying spectra in $\mathcal{C}_*$ near a^0 ? The fourth moment exposes the reference, but its deficits for two unknown spectra can be nearly equal. The anchored certificate therefore does not automatically give a pairwise inverse theorem. One should distinguish an estimator's uniform risk on a neighborhood from confidence-set contraction at its extremal point.

Research question 4 (Ratio profiles without uniform scale spacing). Classify all-error inverse moduli for relative-ratio cones when $b_{j+1}/b_j \rightarrow 0$ or $b_{j+1}/b_j \rightarrow 1$. The algebraic upper certificate survives far beyond (8.1), but its rate need not be attained at every error level by the sparse tail scales. A profile-dependent modulus may retain more information than a single Hölder exponent.

Research question 5 (Exact support and variance with supercritical slack). At the exact boundary the maximal-support fibre is a singleton. With a fixed ratio allowance $\rho > 1/2$, can the predecessor's very small law perturbations be made to preserve both variance and the exact support endpoint? The present rigidity theorem identifies why the critical version is impossible, but does not settle the construction under slack.

Research question 6 (Other factor laws and Gaussian nuisance components). For which centered base laws does a fourth-cumulant extremizer produce the same square-root inverse law? The sign and nonvanishing of the fourth cumulant matter for moment exposure. An additional unknown Gaussian variance changes the second-moment normalization while leaving the fourth cumulant unchanged; separate identifiability and stability statements are needed rather than an unqualified transfer of the present theorem.

Research question 7 (Sharper statistical constants and robust measurements). Determine local testing constants through the leading square-root-density perturbation, and compare the fourth-moment test with a likelihood-based test. Extend the confidence certificate to validated moment intervals, bounded measurement error, or controlled contamination. Such extensions must keep track of the exact structural constraint; an unverified ratio assumption is not supplied by a good empirical fit to four moments.

Research question 8 (A formalized moment-exposure library). Formalize the weighted identity (8.4) and its quantitative consequences as a reusable library for monotone relative-ratio cones. A useful milestone is a kernel-checked proof of the dyadic constant in (1.4), followed by the support singleton theorem. The Hellinger boundary module is a separate, more analytic milestone.

12 Conclusion

The exact dyadic separation boundary is not another instance of the fixed-slack instability. It has a rigid moment geometry. A fourth-moment deficit controls the squared error of every coefficient, producing a sharp square-root inverse law for the entire infinite spectrum and a sharp geometric conditioning law for finite prefixes. Smooth deleted-factor witnesses show that these estimates cannot be improved in exponent, even under the predecessor's explicit derivative bounds. A separate Hellinger argument yields the matching quartic testing complexity.

The essential mechanism is not peculiar to a dyadic Fourier zero lattice. It is the one-sided majorization induced by relative-ratio constraints, which also gives rigidity and sharp rates around nongeometric references. The unresolved questions now lie in optimal constants, two-point rather than anchored recovery, and the interaction of small structural slack with small observational error.

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